

Rail Fence

1. Encrypt the message "Enjoy your Journey" using Rail Fence with depth 2 and 3.

• Depth = 2

E		J		Y		O		R		O		R		E	
	N		O		Y		U		J		U		N		Y

∴ Ciphertext = EJYORORE NOYUTUNY

• Depth = 3

E				Y				R				R			
	N		O		Y		U		J		U		N		Y
		J				O				O				E	

∴ Ciphertext = EYRR NOYUTUNY JOOE

2. Encrypt the message as per your choice using Rail Fence with depth 2 and 3.

• Depth = 2 , Plaintext = CRYPTOLOGY

C		Y		T		L		G	
	R		P		O		O		Y

∴ Ciphertext = CYTLG RPOOY

• Depth = 3, Plaintext = CRYPTOLOGY

C				T				G	
	R		P		O		O		Y
		Y				L			

∴ Ciphertext = C T O G R P O O Y Y L

3. Decrypt the cipher "EOCSNYUWLEJYREASONS", encrypted using Rail fence with depth 4.

Cipher Text length = 19

Depth = 4

*E						*O						*C						*S
	*N				*Y		*U				*W		*L					*E
		*J		*Y				*R		*E				*A		*S		
			*O						*N						*S			

∴ Plaintext = ENJOY YOUR NEW CLASSES.

4. Decrypt the cipher "BODIGUYUBOSNUUTYRKAS", encrypted using Rail fence with depth 3.

Cipher Text length = 20

Depth = 3.

*B				*O				*O				*I				*G			
	*U		*Y		*U		*B		*O		*S		*N		*U		*U		*T
		*Y				*R				*K				*A				*S	

∴ Plaintext = BUY YOUR BOOKS IN AUGUST

5. Decrypt the cipher "SEMSERTISINCMOO" encrypted using Rail fence with depth 2.

Cipher Text length = 15

Depth = 2

*S		*E		*M		*S		*E		*R		*T		*I
	*S		*I		*N		*C		*M		*O		*O	

∴ Plaintext = SSEIMNSCEMRTOIJ

Permutation

1. Encrypt the following message "we are discovered" using the

key "53124"

1	2	3	4	5
5	3	1	2	4
W	E	A	R	E
D	I	S	C	O
V	E	R	E	D

Order : 3 → 4 → 2 → 5 → 1

∴ Ciphertext = ASR RCE EIE EOD WDV

DES

Initial Permutation							
58	50	42	34	26	18	10	02
60	52	44	36	28	20	12	04
62	54	46	38	30	22	14	06
64	56	48	40	32	24	16	08
57	49	41	33	25	17	09	01
59	51	43	35	27	19	11	03
61	53	45	37	29	21	13	05
63	55	47	39	31	23	15	07

1. Using the initial permutation table, determine the output of the initial permutation box when the input is given in hexadecimal as : 0X0002 0000 0000 0001

→ Converting it to binary

00000000 00000010 00000000 00000000 00000000 00000000
00000000 00000001

There are two 1's at bit position 15 and bit position 64.

Applying the Initial Permutation table,

64 is in position 25 at row 4, column 1

15 is in position 63 at row 8, column 7

All other positions are 0.

∴ The output bits are :

00000000 00000000 00000000 10000000 00000000 00000000
00000000 00000010

Converting to hexadecimal

0X0000 0080 0000 0002

2. Given the 4-bit binary Plaintext (M) 1101 and the key (K) 1011, calculate the resulting Ciphertext (C) using a bitwise Exclusive-OR (XOR) operation, show your bit-by-bit work.

• Plaintext (M) : 1101 (binary)

• Key (K) : (1011) binary.

$$\begin{array}{r} 1101 \\ \oplus 1011 \\ \hline 0110 \end{array}$$

∴ Ciphertext (C) = 0110

3. The input to S-box 1 is 000000. What is the output?

S1

14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

000000

To find row, take first and last bits = 00 = 0 (decimal)

Row = 0

To find column, take the middle four bits = 0000 = 0 (decimal)

Column = 0

0th row and 0th column, value is 14

14 in binary 1110

∴ Output = 1110

Product Cipher

Explore ADFGVX Cipher.

• The ADFGVX cipher is a classical encryption technique that combines substitution and transposition.

• The name ADFGVX comes from the six letters,

A, D, F, G, V, X

• These six letters are used to represent the rows and columns of a 6x6 Polybius square.

• Each character is represented by two letters from A, D, F, G, V, X.

• Plaintext → Polybius Substitution → Columnar Transposition → Ciphertext.

	A	D	F	G	V	X
A	A	B	C	D	E	F
D	G	H	I	J	K	L
F	M	N	O	P	Q	R
G	S	T	U	V	W	X
V	Y	Z	0	1	2	3
X	4	5	6	7	8	9

Example: Encrypt "HELLO"

- H → DD
- E → AV
- L → DX
- L → DX
- O → FF

Substitution result: DDAVDXDXFF

For columnar transposition, consider key = CIPHER

1	4	5	3	2	6
C	I	P	H	E	R
D	D	A	V	D	X
D	X	F	F		

∴ ciphertext = DDDVFDXAFX